

1. Introduction

The relationship between competitive advantage and international transferability remains a contested problem in strategy research. Firms that dominate their home markets frequently find capabilities underpinning their dominance erode when moved into host environments that do not reproduce the same conditions in which they were built (Rugman and Verbeke, 2001; Ghemawat, 2001). The arrival of Chinese electric vehicle manufacturers in the European market brings this problem into direct focus. Chinese manufacturers were competing at prices around €10,000 below comparable alternatives (More, 2026). The European Commission attributed this price advantage to state subsidies and industrial policy instead of superior productivity, responding with countervailing duties of up to 35.3% on Chinese-manufactured electric vehicles (EV) on top of the existing 10% standard tariff (Reuters, 2026; Bickenbach et al., 2024). Whether that institutional judgement is correct is a question existing literature has not yet answered. BYD is the world's largest electric vehicle (EV) manufacturer by volume (Richter, 2025), making it a worthy case study: can a firm of this scale and resource depth transfer its domestic competitive advantage into a market which actively works against it. Germany, the largest automotive market in Europe, provides the focal context for that test (GTAI, 2025). This study examines whether BYD's domestic competitive advantage remains transferable in light of European protectionist measures.

The study contributes to both practitioner and academic research. For practitioners, it provides a basis for assessing the strategic threat against European manufacturers: if the tariff succeeds in eroding BYD's competitive advantage, incumbents will have room to compete and innovate, and vice versa. For academic research, Rugman and Verbeke (2001) would predict that, given the scale of Chinese state support and demand-side conditions, BYD's advantage is largely country-specific and should erode substantially once those conditions are removed. If the findings contradict that prediction, they will serve as empirical evidence against it.

2. Literature Review

The EV market refers to the automotive industry segment which specialises in producing, selling and developing EVs (Jagani et al., 2024). The emergence of Chinese manufacturers in western markets has been supported by favourable home-country conditions: infrastructural investment, state intervention and cultural adoption drivers. Within these conditions, BYD has developed substantial cost and battery-technology advantages. The question lies in whether these advantages are substantial enough to challenge the barriers surrounding the transferability of competitive advantage.

2.1 Foundations of BYD Domestic Competitive Advantage

Heffner et al. 's (2007) study on the owners of hybrid-electric vehicles (HEV) concluded that adoption is not only shaped by economic considerations, but instead by the symbolic meaning attached to the technology. The theory's applicability here is partial; the study is almost two decades old, drawn from a US sample concerning HEV's rather than EV's. Its relevance is therefore logical rather than evidential. The symbolism argument made within the study is

assumed to be stronger for EV's, representing a cleaner break from combustion. However, the specific findings cannot be extended to China or Europe without qualification. Qian and Yin (2017) partially fill this gap, linking Chinese EV adoption to culturally embedded values such as Confucianism and Taoism, providing an explanation to the demand-side conditions under which BYD was able to scale domestically.

Funke et al. (2019) identifies charging infrastructure as a critical determinant of EV diffusion. China has the highest share of DC fast chargers globally (C. 40%), reflecting the feasibility constraint imposed by high-density megacity housing, where home charging is uncommon. Ou et al. (2020) documents that, in the same period, the amount of chargers that the United States deployed stood at roughly one-quarter of China's. Both sources are now six years old, but the asymmetry has only widened since; China has only extended its global share of chargers for light-duty vehicles, standing at 70% in 2023 (IEA, 2024). This confirms that the infrastructure-driven divergence in EV architecture has deepened rather than eroded.

State intervention is another factor that facilitated BYD's domestic competitive advantage. The dual-credit system required automakers to generate sufficient new energy vehicle (NEV) credits to offset negative corporate average fuel consumption (CAFC) credits from combustion vehicle sales, and has been progressively tightened since production. Bickenbach et al, (2024) extends this picture beyond credits, identifying favourable taxation and financing as further enablers of domestic scaling.

Porter (1990) and Aghion et al. (2005) argue the importance of market structure in shaping competitive advantage. Intense domestic rivalries drive innovation by reducing complacency and forcing continuous improvement. Ou et al. (2020) documents this pattern in the Chinese EV market, where a wave of new entrants without prior combustion presence compete in a wholly new industry rather than extending established rivalries; unlike the contemporaneous US market, which remained dominated by incumbent automakers with the exception of Tesla

2.2 Theoretical Perspectives on Competitive Advantage

Porter's (1985) framework identifies cost leadership, differentiation and focus as three generic sources of competitive advantage, while also emphasising the importance of vertical integration as a strategic mechanism for realising cost leadership through supply chain control. Under this lens BYD is an archetypal case: long-term lithium partnerships, mineral rights in Brazil's Jequitinhonha Valley and heavy refining investment collectively secure input supply and compress per-unit costs (EVboosters, 2024). In contrast, Tesla, which remains dependent on multiple external suppliers, has flagged lithium shortages as a strategic risk (Tesla, 2024), a vulnerability BYD's integration explicitly reduces by securing input supply and shielding production costs from commodity price volatility.

Barney's (1991) resource-based view provides a different but complementary framework for explaining sustained advantage. The RBV argues that resources must be valuable, rare, inimitable and non-substitutable if they are to generate durable competitive advantage. BYD's pre-automotive battery heritage points to such a resource base, where the firm became one of the first Chinese lithium-ion suppliers to Motorola in 2000 and Nokia in 2002, before acquiring Qinchuan Automotive in 2003 (Zhang, 2016). The Blade Battery represents a tangible outcome of

these accumulated capabilities, and its performance is therefore examined as a key indicator of BYD's resource-based advantage.

Porter and Barney provide recognised benchmarks for identifying what type of advantage BYD appears to possess. Porter allows the analysis to classify BYD's price position as a potential cost-leadership advantage, while Barney provides a basis for treating battery capability as a firm-specific resource. The use of these theories for classification is supported by Dess and Davis (1984) and Newbert (2008). Dess and Davis (1984) use Porter's generic strategies to classify strategic group membership and examine performance differences. Newbert (2008) supports this classificatory use specifically, finding empirically that value and rareness, Barney uses to identify advantageous resources, are reliably associated with competitive advantage at the conceptual level

However, both frameworks are less complete when used to explain whether advantage transfers across national markets. Porter's (1996) subsequent work acknowledges that competitive advantage depends on the strategic fit with the competitive environment and broader strategic context, implying that advantages identified within one market context may not hold in different contexts. Within the RBV, Priem and Butler (2001) identify a parallel constraint; the concept of resource value is relative to the competitive environment, meaning the value of a resource is not intrinsic and instead contingent on the competitive context in which it is deployed. Its strategic worth, therefore, depends on whether the host market rewards its specific outputs.

Kraaijenbrink et al. (2010) and Miller and Dess (1993) extend this critique, arguing that the RBV systematically underplays the extent to which external conditions determine whether internal capabilities translate into advantage. This limitation is central to BYD's case. A cost advantage or battery capability may be identifiable in China, but its strategic value depends on whether the German market rewards the same attributes. Porter and Barney are therefore appropriate for classification; but by their own theoretical logic, cannot fully account for the conditions under which that advantage transfers.

2.3 Limits on the International Transferability of Competitive Advantage

Rugman and Verbeke (2001) challenge the assumption that firms can operate as seamless global competitors, drawing a distinction between firm-specific advantages (FSAs), internal capabilities that are transferable across markets and country-specific advantages (CSAs), which are embedded in home-country conditions such as subsidies, industrial policy, and regulatory support. Their central argument is that many apparent FSAs are in fact CSA-dependent, eroding when the host environment does not reproduce the conditions that originally sustained them. Ghemawat (2001) reinforces this position through the CAGE framework, demonstrating that geographic, administrative, cultural, and economic distances continue to impose meaningful frictions on internationally expanding firms.

Although both studies are now over two decades old, the practical realities they describe have not softened, and if anything, recent protectionist policy suggests intensification. The European tariff operationalises both theories in a single contemporary event. In Ghemawat's terms it represents administrative distance; Rugman and Verbeke's framework would predict it should

erode BYD's cost position by removing the state-derived conditions supporting it, a prediction this study aims to test.

2.4 Research Gap and Hypotheses

While existing literature explains both the emergence of competitive advantages within China's EV ecosystem and BYD's firm-level capabilities, little research has examined whether these advantages translate into equivalent competitiveness in markets operating under the post-2024 tariff regime. Consumer-preference research, though extensive, has rarely been applied to direct cross-model comparison under these conditions. To address this gap, the following hypotheses were derived from the literature review:

H1: BYD models achieve a higher overall competitiveness score than comparable EV models in the German market, as measured by the composite index.

H2: BYD models demonstrate a relative price advantage over comparable EV models in the German market, as measured by the price component of the index.

H3: BYD models demonstrate a relative range advantage over comparable EV models in the German market, as measured by the range component of the index.

The reader may question why the hypotheses anticipate advantage, given theoretical barriers (see sections 2.2 and 2.3). As of early 2026, BYD is the world's largest electric vehicle manufacturer, spanning 112 different countries (Richter, 2025). A firm of this magnitude represents, on Porter's and Barney's own terms, the closest real-world approximation to a resource base substantial enough to overcome the very barriers the literature predicts. H1, H2 and H3 therefore do not ignore the transferability problem; they test whether BYD is the exception that proves the rule.

3. Methodology

3.1 Research Philosophy and Design

This study adopts a positivist philosophical stance, treating competitive advantage as an observable phenomenon that can be operationalised through measurable vehicle attributes rather than a socially constructed perception (Saunders et al., 2019). The research design is deductive and quantitative in nature: hypothesis derived from Porter (1985), Barney (1991) and Rugman and Verbeke (2001) in section 2 are translated into a falsifiable index, which is tested against empirical data derived from KBA registration statistics for the 2025 German EV market. A cross-sectional snapshot is preferred over a longitudinal design, as the research question concerns BYD's current position under the countervailing tariff; a longitudinal design falls outside this scope.

3.2 Sampling Strategy

A purposive sampling strategy was employed in order to select a population whose interests were defined by theory rather than a statistical criteria (Saunders et al., 2019). BYD's four highest-volume models were identified from KBA registration data: Dolphin Surf, Atto 2, Seal and Sealion 7. The selection represents vehicles genuinely competitive within the market rather than peripheral entries. For each BYD model, a segment-specific set of comparators was constructed using five criteria, applied identically across all four segments to preclude section bias:

Vehicle Type: All competitors had to be fully electric and share the body type of each respective BYD model.

Boot Space had to fall within 30% of the BYD model (proxy for vehicle class)

Battery capacity had to fall within 20% to keep the energy platform consistent.

Price had to fall within 35%, the percentage is deliberately broad to include higher-priced incumbents, avoiding the exclusion of BYD's cost position.

Sales Volume had to fall within 20% of the lower band of BYD's 2025 registrations, ensuring that comparable models had demonstrated market relevance rather than competitiveness based only on specification.

Before the final thresholds were selected, pilot indices were constructed. The final criteria was retained because they produced the strongest balance between comparator volume and market realism. The current benchmarks therefore reflect values that are narrow enough to exclude weak substitutes, but broad enough to capture the vehicles German consumers could plausibly consider alongside each BYD model.

3.3 Data Sources

Vehicle specifications (price, range, charge time, efficiency)¹ were collected from manufacturer technical datasheets and cross-validated across EV-database listings to mitigate single-source bias.

3.4 Index Construction and Weighting

Each variable was standardised via 0-1 min-max normalisation, eliminating distortion that would otherwise arise from heterogeneous units (€, KM, minutes, Wh/Km). Normalised scores are aggregated using weights derived from published practitioner and academic research on EV purchase preferences. (Zayer, et al., 2025; Venus et al., 2024; Jagani et al., 2024). Battery warranty and depreciation were considered given the importance of these factors to consumers, but ultimately excluded. Insufficient variation between battery warranties and thin secondary market presence for recently launched models would have introduced inaccuracy rather than accurate signals.

3.5 Analytical Strategy

A composite index was chosen over regression specification as the research is cross-sectional and comparative in nature. The study is concerned with the relative positioning of BYD against identifiable rivals, rather than estimating causal relationships between variables - a defining factor of regression. BYD's position within each segment is therefore assessed through four

¹ Listed in order of importance to consumers

complementary descriptive measures: (i) **rank** within the segment; (ii) **deviation from the segment mean**; (iii) **Z-score** standardisation of the deviation against segment dispersion, distinguishing genuine outperformance from an artefact of segment clustering; and (iv) **quartile placement** (below Q1/ within IQR / above Q3). Using the four lenses in combination mitigates any risk of a single metric misrepresenting BYD's position.

3.6 Methodological Limitations

Three limitations merit disclosure. First, the index presupposes consumer rationality and therefore omits the idea of perception, dealer density and after sales network, non-trivial drivers of German EV choice. Secondly, the cross-sectional design will not capture post-tariff convergence, a longitudinal panel across 2024–2027 registrations would have directly tested whether BYD's position is narrowing or widening, though such a design only becomes feasible from 2026 onward, when sufficient post-tariff registration data becomes available. Finally, constructing a meaningful comparator base introduced selection pressures that may have distorted the sample. Price and battery capacity thresholds were deliberately broadened to ensure sufficient competitor volume, which risks drawing in vehicles outside BYD's realistic competitive set. The sales volume criterion compounds this, as recently launched models that BYD may directly compete with in practice were excluded on the grounds of insufficient registration data.

4. Analysis

This section presents the results of the composite index across all four segments, followed by the isolated tests of BYD's price (H2) and range (H3) components.

4.1 Overall Competitive Position (H1)

The following information can be found in *Tables 1-4*

There is a degree of consistency in three of the four segments. Two models, the Dolphin Surf and Sealion 7, fall more than one standard deviation below the mean, placing beneath the 25th percentile of their respective distributions. The Dolphin Surf records the lowest score in its entire segment (0.30 against a segment maximum of 0.67), and the Sealion 7 is outperformed by 10 of its eleven competitors, including the incumbent European brands (Audi Q4, Skoda Enyaq and VW ID.4) and direct Chinese rivals (X-Peng G6 and Polestar 4).

The Atto 2 segment exhibits materially lower dispersion ($\sigma = 0.083$) than the other three segments, with all ten models clustered between 0.42 and 0.50 once the BMW X1 outlier is set aside. BYD's Z-score of +0.06 is therefore not materially different from the mean, therefore, the Atto 2 is neither advantaged nor disadvantaged within a segment where differentiation on the measured variables is limited.

The Seal is the only BYD model where the aggregated result admits a more promising reading. At 0.49, it sits 0.02 below the segment mean, but the segments high dispersion ($\sigma = 0.22$) reflects two sub-populations: premium European saloons underperforming on price (BMW 4ER 0.22, Mercedes EQE 0.25) and technology-led rivals dominating the top tier (Tesla Model 3 0.81,

Polestar 2 0.66, Hyundai 6 0.65). BYD's mid-table position reflects the high variability within the segment, rather than clear underperformance.

Taken as a whole, H1 is not supported by the data. BYD holds a defensible position in only one of four segments, and in two segments its aggregate score is more than a full standard deviation below the segment mean.

4.2 Price Competitiveness (H2)

The following information can be found in *Tables 5-8*

Isolating the price component yields a pattern distinct from the composite index tested in H1. The results do not support a consistent price advantage. The Seal is the clearest case, ranking second in its segment with a price score of 0.34 against a segment mean of 0.225, outperforming every European saloon in the sample on price. This is the only segment where BYD demonstrates a price position that can be characterised as a genuine advantage.

The Atto 2 and Sealion 7 sit at or immediately adjacent to their respective segment means ($Z = +0.01$ and $Z = -0.07$ respectively), indicating price parity rather than advantage. Despite the embedded EU countervailing tariff, BYD has absorbed the duty sufficiently to avoid being priced out of these segments, but has not retained a meaningful cost edge over its comparators. The Dolphin Surf is the weakest case, scoring 0.11 against a segment mean of 0.167 ($Z = -0.41$), falling below four of its five comparators including the Leapmotor T03, a direct Chinese rival operating under the same tariff regime.

On balance, H2 is not supported. BYD retains price parity in two segments and a genuine price advantage in one, but falls below the segment mean in a fourth. The tariff has compressed whatever cost advantage once existed to the point where it is indistinguishable from average market pricing in most segments. The Seal provides partial support for the cost-leadership claim, but three models at-or-below the segment mean does not constitute a consistent price advantage across the range.

4.3 Range Performance (H3)

The following information can be found in *Tables 9-12*

The range component produces the most consistent pattern in the study. Every BYD model scores below the segment mean on range, and three of the four sit at, or near the segment minimum. The Sealion 7 recorded a normalised score of 0.0, the lowest possible score in its segment where the Model Y and Ioniq 5 each scored 0.30.

Notably, apart from polestar models, the other Chinese competitor models in the sample - Leapmotor T03, MG 4, Smart #1 and X-Peng G6 - all record ranges scored below their respective means. Interestingly, all of these models use Lithium Iron-Phosphate (LFP) batteries over Nickel Manganese Cobalt (NMC), and other non-chinese models using LFP technology also exhibited sub-par range scores. The range disadvantage is therefore not isolated to BYD: it is a feature of vehicles built on LFP architectures, which comprise the majority of Chinese manufactured EVs.

As such, H3 is rejected, but carries a nuance the hypothesis itself did not anticipate. The data does not demonstrate a firm-specific battery weakness at BYD; it demonstrates a structural range disadvantage common to the battery chemistry class that BYD has chosen. Whether that choice reflects a deliberate trade-off or a constraint is a question for Section 5.

4.4 Insights in Response to the Research Question

The research question was concerned with whether BYD's domestic competitive advantage is transferable to the German market. The composite index provides little empirical support, with BYD underperforming in three of four segments. Disaggregating the index reveals that the results are neither uniform nor entirely attributable to BYD itself. Price parity is maintained in two segments and a genuine price advantage exists in one, suggesting a residual cost capability survives the tariff, though not at a level that constitutes a consistent competitive advantage. Range underperformance is systematic across every LFP-equipped vehicle in the sample, indicating the deficit is attributable to battery chemistry rather than firm-level deficiency.

5. Discussion

This section interprets the pattern of results across all four segments and considers their implications for the wider strategy literature.

5.1 Competitive Advantage and Market Fit

The rejection of H1 is broadly consistent with Rugman and Verbeke's (2001) prediction that firm-specific advantages weaken when removed from the country's conditions that produced them, though the data identifies a more precise explanation. Seven of China's best-selling EVs in H1 2025 were micro or budget city cars (Autovista24, 2025), a category largely absent from the German market, where demand is centred around mid-size crossovers and family vehicles (R&M, 2025). The Dolphin Surf illustrates this directly: sold domestically as the Seagull, BYD's highest-volume model, it required up-specification before European launch (McNamara, 2025). The issue is not a lack of capability, it is an example of the misalignment between Chinese and German consumers. This refines Rugman and Verbeke's framework: domestic advantages hinge on the national conditions in which they were built and the product category the conditions reward.

H2's findings are the most theoretically significant in the study. Rather than imposing quotas, the EU applied countervailing tariffs that directly targeted price leadership from Chinese brands, created by state support, scale of domestic demand and supply-chain integration. The data shows that in two of four segments, the Atto 2 and Sealion 7, BYD's price position is statistically indistinguishable from the segment mean. This is not a price advantage; it is the absence of a price disadvantage. The fact that BYD has absorbed the tariff burden without being priced out of these segments is consistent with Porter's (1985) argument that vertical integration generates genuine production-cost efficiency, but the claim goes no further: parity is not advantage, and the data does not support the inference that BYD holds a Porterian cost-leadership position in these segments under the current tariff regime. The Seal provides the strongest evidence in favour of H2, ranking second in its segment on price and outperforming every European saloon

in the sample. From a Rugman and Verbeke (2001) perspective, this refines the CSA-dependency claim. The partial retention of a price position, despite a tariff designed to strip out country-specific pricing advantages, suggests the residual advantage in this segment is at least partly firm-specific rather than purely policy-derived. The Dolphin Surf is the counter-case, where European specification upgrades eroded the price cushion entirely, combining the CSA-level product-category mismatch identified in 5.1 with FSA erosion through higher production costs. Every other model faces only one of these constraints; the Dolphin Surf faces both.

The rejection of H3 identifies a flaw in the hypothesis rather than BYD's capability. LFP chemistry is engineered for safety, cycle longevity and cost, not energy density (Evro et al., 2024), and every LFP-equipped vehicle in the sample falls in the lower half of its segment's range scores regardless of manufacturer. BYD's own messaging confirms this; the Blade Battery is positioned around safety and packaging, not range (BYD, 2025). The hypothesis incorrectly equated battery capability with range advantage rather than the attributes LFP is designed to deliver. More broadly, this illustrates Priem and Butler's (2001) point that a capability is only advantageous where the host market rewards its specific output. LFP suits China's dense fast-charging environment but is less suited to Germany, where home charging and single-charge range dominate consumer priorities. H2 and H3 read together reveal a coherent pattern: price parity partly offsets the range deficit in most segments, but in the Dolphin Surf segment, where adaptation has eroded even that parity, no such compensation exists.

5.2 Contributions to wider literature

The study is able to connect the findings to three wider areas of strategy literature: strategic positioning and trade-offs, where cost leadership appears alongside weaker range and product-market fit, suggesting advantage cannot be optimised across all dimensions simultaneously (Porter, 1996); core competence, where BYD's battery and integration capabilities remain valuable only where the market rewards their specific outputs, illustrating that competencies do not transfer uniformly (Prahalad and Hamel, 1990); and international adaptation, where Germany demonstrates the limitations of applying a successful home-market model without adjustment to local demand conditions (Levitt, 1983; Yip, 1989).

6. Conclusion

This study concludes that BYD's domestic competitive advantage does not transfer to the German EV market as predicted. However, price parity suggests a degree of transferability; BYD avoids being priced out of two segments and retains a genuine price advantage in one, the Seal, despite the additional EU tariff. This supports the view that BYD's vertical integration and production scale contain a genuine firm-specific component. This does not, however, translate into an overall competitive advantage. The rejection of H1 and H3 shows that BYD underperforms on the composite measure in three of four segments, with range performance weakened by the LFP battery strategy and its product offering less closely aligned with German demand for larger crossovers, SUVs and family-oriented EVs than the urban and budget categories that helped BYD scale in China.

The wider implication is that competitive advantage is context-dependent. Porter and Barney help define BYD's cost and resource advantages, but their own theoretical frameworks acknowledge that advantage is contingent on fit with the competitive environment, a fit that does not fully hold in Germany. Rugman and Verbeke therefore provide the stronger explanatory frame: BYD's advantage is neither purely firm-specific nor purely country-specific, but a combination of both, and the tariff exposes the boundary between them.

The study carries two primary limitations: its reliance on a small purposive sample, and its use of specification-based measures rather than realised market outcomes. The latter is partially mitigated by the sample's direct relevance to the segments in which German EV demand is currently concentrated, and where Chinese entrants face the most direct competitive pressure. Future research could strengthen the analysis by incorporating sales data and depreciation costs once Chinese EV entrants have accumulated sufficient market presence to generate reliable secondary market data. A longitudinal design would be particularly valuable, as it would allow researchers to assess how BYD's competitive position evolves as the German EV market matures, consumer familiarity grows and Chinese brands become an established part of the competitive landscape rather than new entrants.

7. References

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8. Appendix

Table 1

Segment	n	Mean	SD	Min	Q1	Median	Q3	Max	IQR	BYD Score	BYD Rank	BYD Δ Mean	BYD Z	BYD Quartile
Dolphin Surf	6	0.475	0.141	0.30	0.373	0.470	0.568	0.67	0.195	0.30	6/6	−0.18	−1.24σ	Q1
Atto 2	10	0.435	0.086	0.20	0.433	0.460	0.475	0.50	0.043	0.44	7/10	+0.01	+0.06σ	Q2
Sealion 7	12	0.443	0.104	0.30	0.360	0.455	0.478	0.62	0.118	0.32	11/12	−0.12	−1.18σ	Q1
Seal	6	0.513	0.238	0.22	0.310	0.570	0.658	0.81	0.348	0.49	4/6	−0.02	−0.10σ	Q2

Table 2

Rank	Make	Model	Price	Charge	Range	Efficiency	Index	Δ Mean	Z-Score	Quartile
1	Hyundai	INSTER	0.17	0.20	0.30	0.00	0.67	+0.20	+1.38	Q4
2	Citroën	C3	0.24	0.13	0.12	0.10	0.59	+0.12	+0.82	Q3
3	Opel	Corsa	0.00	0.20	0.26	0.04	0.50	+0.03	+0.18	Q2
4	Leapmotor	T03	0.40	0.00	0.00	0.04	0.44	−0.04	−0.25	Q2
5	Renault	5	0.08	0.13	0.13	0.01	0.35	−0.13	−0.89	Q1
6	BYD	Dolphin Surf	0.11	0.14	0.01	0.04	0.30	−0.18	−1.24	Q1

Table 3

Rank	Make	Model	Price	Charge	Range	Efficiency	Index	Δ Mean	Z-Score	Quartile
1	Hyundai	IONIQ 5	0.19	0.20	0.19	0.04	0.62	+0.18	+1.71	Q4
2	Tesla	Model Y	0.20	0.11	0.30	0.00	0.61	+0.17	+1.61	Q4
3	Kia	EV6	0.15	0.16	0.14	0.05	0.50	+0.06	+0.55	Q4
4	Polestar	4	0.00	0.11	0.30	0.07	0.47	+0.03	+0.26	Q3
5	VW	ID.4 / ID.5	0.23	0.10	0.12	0.03	0.47	+0.03	+0.26	Q3
6	X-Peng	G6	0.19	0.15	0.06	0.06	0.46	+0.02	+0.16	Q3
7	Skoda	ENYAQ	0.19	0.11	0.13	0.02	0.45	+0.01	+0.07	Q2
8	Toyota	bZ4X	0.20	0.09	0.07	0.05	0.42	−0.02	−0.22	Q2
9	Audi	Q4	0.13	0.07	0.09	0.08	0.37	−0.07	−0.70	Q2
10	Opel	GRANDLAND X	0.19	0.00	0.04	0.10	0.33	−0.11	−1.09	Q1
11	BYD	Sealion 7	0.15	0.09	0.00	0.08	0.32	−0.12	−1.18	Q1
12	Cupra	TAVASCAN	0.04	0.13	0.08	0.05	0.30	−0.14	−1.38	Q1

Table 4

Rank	Make	Model	Price	Charge	Range	Efficiency	Index	Δ Mean	Z-Score	Quartile
1	Tesla	Model 3	0.40	0.11	0.30	0.00	0.81	+0.30	+1.25	Q4
2	Polestar	2	0.27	0.00	0.30	0.08	0.66	+0.15	+0.62	Q3
3	Hyundai	IONIQ 6	0.22	0.20	0.20	0.03	0.65	+0.14	+0.57	Q2
4	BYD	Seal	0.34	0.02	0.04	0.10	0.49	−0.02	−0.10	Q2
5	Mercedes	EQE	0.00	0.04	0.17	0.09	0.25	−0.26	−1.10	Q1
6	BMW	i4	0.12	0.04	0.00	0.05	0.22	−0.29	−1.23	Q1

Table 5

Rank	Make	Model	Price Score	Δ Mean	Z-Score	Quartile
1	Leapmotor	T03	0.40	+0.23	+1.66	Q4
2	Citroën	C3	0.24	+0.07	+0.52	Q3
3	Hyundai	INSTER	0.17	+0.00	+0.02	Q2
4	BYD	Dolphin Surf	0.11	−0.06	−0.41	Q2
5	Renault	5	0.08	−0.09	−0.62	Q1
6	Opel	Corsa	0.00	−0.17	−1.19	Q1

Table 6

Rank	Make	Model	Price Score	Δ Mean	Z-Score	Quartile
1	Opel	FRONTERA	0.40	+0.16	+1.37	Q4
2	Kia	EV3	0.33	+0.09	+0.77	Q4
3	Skoda	ELROQ	0.29	+0.05	+0.43	Q3
4	MG	4	0.27	+0.03	+0.26	Q3
5	Ford	EXPLORER	0.27	+0.03	+0.26	Q3
6	Smart	#1	0.27	+0.03	+0.26	Q2
7	Hyundai	KONA	0.25	+0.01	+0.09	Q2
8	BYD	Atto 2	0.24	+0.00	+0.01	Q1
9	Mercedes	EQA	0.07	−0.17	−1.44	Q1
10	BMW	iX1	0.00	−0.24	−2.02	Q1

Table 7

Rank	Make	Model	Price Score	Δ Mean	Z-Score	Quartile
1	VW	ID.4 / ID.5	0.23	+0.08	+1.09	Q4
2	Tesla	Model Y	0.20	+0.05	+0.65	Q4
3	Toyota	bZ4X	0.20	+0.05	+0.65	Q4
4	Hyundai	IONIQ 5	0.19	+0.04	+0.51	Q3
5	X-Peng	G6	0.19	+0.04	+0.51	Q3
6	Skoda	ENYAQ	0.19	+0.04	+0.51	Q3
7	Opel	GRANDLAND X	0.19	+0.04	+0.51	Q2
8	Kia	EV6	0.15	−0.01	−0.07	Q2
9	BYD	Sealion 7	0.15	−0.01	−0.07	Q2
10	Audi	Q4	0.13	−0.03	−0.36	Q1
11	Cupra	TAVASCAN	0.04	−0.12	−1.67	Q1
12	Polestar	4	0.00	−0.16	−2.24	Q1

Table 8

Rank	Make	Model	Price Score	Δ Mean	Z-Score	Quartile
1	Tesla	Model 3	0.40	+0.18	+1.19	Q4
2	BYD	Seal	0.34	+0.12	+0.78	Q3
3	Polestar	2	0.27	+0.05	+0.31	Q2
4	Hyundai	IONIQ 6	0.22	−0.01	−0.04	Q2
5	BMW	i4	0.12	−0.11	−0.72	Q1
6	Mercedes	EQE	0.00	−0.23	−1.53	Q1

Table 9

Rank	Make	Model	Range Score	Δ Mean	Z-Score	Quartile
1	Hyundai	INSTER	0.30	+0.16	+1.32	Q4
2	Opel	Corsa	0.26	+0.12	+0.99	Q3
3	Renault	5	0.13	−0.01	−0.06	Q2
4	Citroën	C3	0.12	−0.02	−0.14	Q2
5	BYD	Dolphin Surf	0.01	−0.13	−1.02	Q1
6	Leapmotor	T03	0.00	−0.14	−1.10	Q1

Table 10

Rank	Make	Model	Range Score	Δ Mean	Z-Score	Quartile
1	Mercedes	EQA	0.30	+0.21	+2.25	Q4
2	Hyundai	KONA	0.20	+0.11	+1.16	Q4
3	BMW	iX1	0.12	+0.03	+0.29	Q3
4	MG	4	0.08	−0.01	−0.14	Q3
5	Ford	EXPLORER	0.07	−0.02	−0.25	Q3
6	Kia	EV3	0.06	−0.03	−0.36	Q2
7	Skoda	ELROQ	0.04	−0.05	−0.58	Q2
8	BYD	Atto 2	0.04	−0.05	−0.58	Q1
9	Smart	#1	0.02	−0.07	−0.77	Q1
10	Opel	FRONTERA	0.00	−0.09	−1.01	Q1

Table 11

Rank	Make	Model	Range Score	Δ Mean	Z-Score	Quartile
1	Tesla	Model Y	0.30	+0.18	+1.91	Q4
2	Polestar	4	0.30	+0.18	+1.91	Q4
3	Hyundai	IONIQ 5	0.19	+0.07	+0.76	Q4
4	Kia	EV6	0.14	+0.02	+0.23	Q3
5	Skoda	ENYAQ	0.13	+0.01	+0.13	Q3
6	VW	ID.4 / ID.5	0.12	+0.00	+0.02	Q3
7	Audi	Q4	0.09	−0.03	−0.30	Q2
8	Cupra	TAVASCAN	0.08	−0.04	−0.40	Q2
9	Toyota	bZ4X	0.07	−0.05	−0.51	Q2
10	X-Peng	G6	0.06	−0.06	−0.61	Q1
11	Opel	GRANDLAND X	0.04	−0.08	−0.81	Q1
12	BYD	Sealion 7	0.00	−0.12	−1.24	Q1

Table 12

Rank	Make	Model	Range Score	Δ Mean	Z-Score	Quartile
1	Tesla	Model 3	0.30	+0.13	+1.04	Q4
2	Polestar	2	0.30	+0.13	+1.04	Q3
3	Hyundai	IONIQ 6	0.20	+0.03	+0.25	Q2
4	Mercedes	EQE	0.17	+0.00	+0.02	Q2
5	BYD	Seal	0.04	−0.13	−1.01	Q1
6	BMW	i4	0.00	−0.17	−1.32	Q1

Note to marker: AI generated tables for formatting purposes